HIDDEN TECH DEBT IN AGENTS

In 2015, Google warned us:

"The model is just a tiny box."

11 years later, the SAME warning hits AI Agents — and 99% of teams miss it.

What production agents really look like →

Akhila
AI & Tech

Follow me

THE REAL JOB

The model is the small box. Everything else is the JOB.

The 9 components where production agents actually live and die.

- Beginner-friendly, with analogies
- Concrete next steps per layer
- The runtime is the new ML infrastructure

Akhila
AI & Tech

Follow me

THE ORIGIN

The 2015 Lesson That Predicted Today

A paper that named our profession

In 2015, Google engineers published a NeurIPS paper: **"Hidden Technical Debt in Machine Learning Systems."** The point was brutal:

- The model code is a rounding error
- Everything around it — pipelines, monitoring, serving — is where systems live, fail & accrue debt

It spawned MLOps

And every "ML Engineer" job title you scroll past today.

The same pattern is happening RIGHT NOW with AI Agents.

Akhila
AI & Tech

Follow me

THE REMIX

02

Same Diagram. New Decade.

"Everything else" got rewritten

In 2015 it was

Feature stores · data pipelines · drift monitoring · model serving.

In 2026 it is

Orchestrator · tool layer · memory · sandbox runtime · tracing · eval harness · guardrails.

The model is rented. Your infrastructure is your IP.

Akhila

AI & Tech

Follow me

COMPONENT #1

Markdown Configs

Your prompts are the real source code

Prompts, skills, sub-agent definitions, slash commands are NOT docs. When you "build an agent," 80% of the work is editing prose: system prompts, tool descriptions, few-shot examples, role specs.

- **Version control every prompt (prompts/main_agent.md)**
- **Diff prompt changes in PR reviews like code**
- **Tag versions & tie them to eval scores**
- **Roll out behind feature flags**

Markdown configs deserve the same discipline as code.

Akhila
AI & Tech

Follow me

04

COMPONENT #2

The Model Layer

"How many models, and which fires when?"

Analogy 12 calls on a frontier model ≈ $0.40. Move 60% to a small model ≈ $0.18. Multiply by 10,000 daily users → a 6-figure annual decision.

A real setup has 3 things

A router (often non-LLM) · a model fleet (small/mid/large + specialists) · a fallback chain for when Provider A is down.

Use LiteLLM / OpenRouter / AI Gateway — or switching = a 6-week project.

Akhila

AI & Tech

05

COMPONENT #3

The Agent Harness

Open any production agent — it's a state machine

ReAct

Reason → Act → Observe → Repeat. Robust, slightly token-heavy. The right default.

Plan-and-Execute

Planner makes the plan, executor runs it. Cheap, but brittle when plans change.

Reflexion / Self-Critique

A critic pass after each step. Expensive, but raises quality on hard tasks.

Guardrails it MUST own: step limit · $ circuit breaker · smart retries.

Akhila
AI & Tech

Follow me

COMPONENT #4

06

The Tool Layer

The agent doesn't see tools — it sees descriptions

Analogy One team's agent burned its whole monthly API budget in an afternoon — it re-called a failing API 47 times. The bug wasn't the API. It was the tool description.

Every tool has 3 parts

Schema (in/out) · implementation (the code) · description (the natural-language hint — the highest-leverage piece).

15 narrow tools beat 1 mega-tool — accuracy collapses at 15–25 tools.

Akhila
AI & Tech

Follow me

COMPONENT #5

07

Memory

Not one vector DB — a stack of subsystems

Tier 1 — Scratchpad

Current conversation + recent tool outputs. Solve: summarization & pruning.

Tier 2 — Session

Rolling summary + last N raw turns. Solve: consistency — this is where bugs live.

Tier 3 — Long-term

Persists across sessions/users. Solve: retrieval without noise.

The 3 AM bug is session summaries drifting from raw history.

Akhila

AI & Tech

COMPONENT #6

08

Serving Infrastructure

An agent run is NOT a web request

Analogy A web request is short, stateless, uniform. An agent run is long, stateful, heterogeneous — and spends most of its life IDLE, waiting on tools, webhooks & human approval.

5 questions your serving design must answer:

- Invocation model — sync, async, or event-triggered?
- Where does state live — in-process or external DB?
- Scaling — fan-out, multi-tenant, long-lived?
- Can serving talk to your tracing?
- Cost — are you billed during idle time?

Akhila
AI & Tech

Follow me

09

COMPONENT #7

Observability & Tracing

Don't look at the answer — look at the 14 steps before it

- Step 7: search returned empty → model fabricated a result
- Step 12: JSON parser silently dropped a field
- Step 4–5: orchestrator looped silently

5 metrics that matter

Task success rate (binary) · cost per successful outcome · P95 latency · tool error rate · avg steps per task.

Tools: LangSmith · Langfuse · W&B Traces · Logfire · Opik.

Akhila
AI & Tech

Follow me

10

COMPONENT #8

The Evaluation Harness

No evals = a demo with vibes attached

4 types you'll need

Reference-based (golden in/out) · reference-free (LLM-as-judge) · trajectory (score the path) · end-to-end (real-user shadow mode).

Beware "eval set rot" — refresh monthly with sampled real traffic.

The eval suite is THE deliverable. The agent just scores well on it.

Akhila

AI & Tech

Follow me

COMPONENT #9

Guardrails & Safety

Threat-model your agent — what can go wrong?

- Prompt injection in the user message
- Prompt injection in tool output
- PII leaking from memory across users
- Runaway cost from malicious 30-step loops
- Off-policy financial / medical / legal advice

3 engineering rules

Rules in code, not prompts · human-in-the-loop on irreversible actions · guardrails are a design constraint, not a last-minute feature.

Defense in depth — Swiss cheese: the holes never line up.

Akhila
AI & Tech

Follow me

THE BIG PICTURE

Your agent stack in 2026

- **Markdown Configs** — the source code
- **Model Layer** — router + fleet
- **Agent Harness** — state machine
- **Tool Layer** — schema + description
- **Memory** — 3 tiers
- **Serving Infra** — 5 design questions
- **Observability** — trace + monitor
- **Eval Harness** — THE deliverable
- **Guardrails** — defense in depth

The model is the small box. Everything else is the JOB.

Akhila
AI & Tech

Follow me

13

TAKEAWAY

If You're Building Agents in 2026

- The model is the small box — don't optimize it first
- The runtime is the new ML infrastructure — pick it deliberately
- Own the whole picture, or own the seams between components

The role has a name

AI Systems Engineer. Same shape as MLOps in 2015 — new decade, new components.

Credit Based on Miguel Otero Pedrido's brilliant article on The Neural Maze — go read the full piece for deeper insights.

Akhila
AI & Tech

Follow me

FOUND THIS USEFUL?

Share it with your team

Repost Save Comment

Follow Akhila for more such insights →

Akhila

AI & Tech

Follow me